

Challenges and Requirements in Multi-Drone Interfaces

Maria-Theresa Oanh Hoang

Aalborg University
Aalborg, Denmark
mtoh@cs.aau.dk

Mikael B. Skov

Aalborg University
Aalborg, Denmark
dubois@cs.aau.dk

Niels van Berkel

Aalborg University
Aalborg, Denmark
nielsvanberkel@cs.aau.dk

Timothy Robert Merritt

Aalborg University
Aalborg, Denmark
merritt@cs.aau.dk

ABSTRACT

Unmanned aerial vehicles (UAVs), commonly known as drones, have been deployed across various applications. These applications range from entertainment to critical situations, such as search and rescue (SAR) operations. The use of single drones is most common—one pilot controls one individual drone. Research has begun to explore the benefits of deploying a group of drones as a coordinated swarm. It is, however, uncertain how a multi-drone system should be designed to facilitate interaction in real-world contexts. We report initial findings from three study sessions involving prototype evaluations and co-design sessions we conducted in collaboration with the emergency services of Denmark. The results of our study open new questions and provide input on the features and functions that impact the future adoption of multi-drone systems, including interactions with multiple video feeds, ecology of screens, team communication, and flight control methods.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design; Empirical studies in ubiquitous and mobile computing; Collaborative and social computing.**

KEYWORDS

search and rescue, multi-drone system, user-interface evaluation

ACM Reference Format:

Maria-Theresa Oanh Hoang, Niels van Berkel, Mikael B. Skov, and Timothy Robert Merritt. 2023. Challenges and Requirements in Multi-Drone Interfaces. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3544549.3585673>

1 INTRODUCTION

In recent years, there has been a rise in the use of drones across various contexts and application areas, often involving direct interaction with people. Drones have become tools for various causes,

including the inspection [7, 39] and mapping of land [19, 41], delivering items [17], and creating intricate light shows [42]. Furthermore, drones are used for critical situations such as emergency response and search and rescue [20].

Traditionally, interactions with drones have focused on a single operator controlling a single drone, often utilizing a phone or remote controller such as a two-stick controller (e.g., DJI's Smart Controller¹). These systems require the pilot to fly the drone by moving the joysticks to adjust the position from moment to moment, which limits the speed at which tasks can be completed. Programmable flight planning helps to automate flight control. However, this is useful only for limited scenarios where the path can be planned ahead of the flight. Both research and industry have started to push the boundaries of automation with intelligent flight planning² [34, 40], as well as expanding beyond the deployment of single drones to multiple drones that can work together, also known as *swarms* [1, 8, 11, 39]. The potential benefits of drone swarms include more rapid visual scanning of large areas and the ability to provide multiple camera perspectives, among others [35].

While contemporary physical controllers are optimized for controlling a single drone by a single pilot, they are not designed to control multiple drones simultaneously. Therefore, multiple drones require multiple pilots, with the coordination between pilots introducing increased complexity and diminishing the situation awareness (SA). Further research is necessary to understand how to design user interfaces to support people interacting with drone swarms to realize the benefits mentioned above [8, 23] while maintaining SA [1, 14, 15, 38].

In this paper, we investigate how a multi-drone system could be designed to specifically support search and rescue (SAR) missions. SAR teams consist of professionals with varying technical expertise and a wide range of experiences they draw from when deploying SAR missions. Due to the critical nature of these missions, the system needs to facilitate effective planning and control of the swarm and facilitate communication and situation awareness to support the SAR team.

Building upon the HCI studies that have previously focused on SAR [5], our work continues in this tradition by directly involving end-users. By doing so, we aim to gain a deeper understanding of the current challenges and design solutions that are specifically tailored to time-sensitive and critical situations. This paper presents current challenges and requirements for multi-drone interfaces through

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '23, April 23–28, 2023, Hamburg, Germany

© 2023 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3585673>

¹<https://www.dji.com/uk/smart-controller>

²<https://www.robotto.ai/fire>

a study comprising three sessions. We created a SAR prototype system capable of controlling multiple drones flying outdoors to elicit a more realistic scenario than a lab study would provide. From our co-design sessions, we identified various requirements relevant to the SAR context but could also apply to other uses of drone swarms.

We provide the following contributions. First, we present example user interfaces that help in the planning and real-time control of a drone swarm. Second, we identify key challenges among the human team interacting and controlling the swarm. Finally, we propose requirements and approaches to designing multi-drone interfaces for future research.

2 INITIAL PROTOTYPE

We developed an initial prototype system, in which we make use of familiar visual cues from the platform currently used by the emergency services. In this section, we describe the hardware and software of the initial prototype used in the first study session. Following Sessions 1 and 2, we added additional features to this system, as described in the respective study sessions.

2.1 Hardware and Software

The drone control prototype consists of three parts: a drone controller, a server, and a web application.

2.1.1 Drones and Smart Controllers. We used drones from *Dajiang Innovations* (DJI). Specifically, we used two **DJI Mavic 2 Enterprise Advanced Dual** and four **DJI Air 2S**, where each drone was connected to a **DJI Smart Controller**. The DJI products were chosen because the emergency services are familiar with them and use them in their drone operations. We installed our custom Android app based on DJI's software development kit on the Smart Controller, which communicates with the drone and the server.

2.1.2 Web Server and Client. We developed a web application to facilitate interaction between drone operators and drones. Implementing it as a web application had the added benefit of being platform-independent and, thereby, could be tested on any device with a browser. To link the web clients with the Smart Controllers, we created a server hosted on a local computer that acted as a tunnel for messages. For each individual drone controller, each instance of the web client continuously listens to changes in the server and updates the layout accordingly.

The interface centers around a map view, following prior work on multi-drone systems [2, 16]. This map shows where the drones are flying and where they are heading. We made an effort to follow design cues from DJI's in-built software since the emergency services already have some experience with it. For example, we implemented a feature that allowed the drone operators to fence in an area that the drones would stay within during the search. We also added a menu to provide the current drone status and additional control methods described in the next section.

2.2 Control Methods

We implemented three different **control methods** for controlling the drones: *selection*, *beacon* and *leader drone* [26, 27, 33]. With *selection control*, the user selects a subset of the active drones and

points them to a specific location on the map. We considered this as an appropriate control method in a location containing something crucial to the mission, e.g., a building or something that might resemble a person. The *beacon control method* has two modes: **repel** and **attract**. This method allows the user to fence off an area of particular interest or should be avoided. With the *leader drone control*, the operator would mainly interact with a designated leader drone, and the swarm would inherit features and behaviors from the designated leader.

3 THREE EVALUATION SESSIONS WITH SAR PROFESSIONALS

Our research is exploratory and involves evaluations of functional prototypes with experts in SAR. Our study approach was inspired by the *say, do, make* framework [32, 36], where we primarily took elements from the *say* and *make* tools and techniques. Users engaged in realistic tasks using the prototypes to evaluate the usability of features and functions, after which we engaged in co-design discussions with pre-defined questions to generate ideas and suggestions for how the future system *should* be designed. We conducted three study sessions in collaboration with the emergency services of Denmark across three cities in the country. In total, six participants joined the study (all male), five of whom are active SAR officers, and one is their IT specialist who acquires and supports the technologies used by the team. Depending on the session, the participants had varying degrees of SAR and drone experience, as summarized in Table 1. Figure 1 illustrates the focus of the different sessions, including the prototype evaluations in the field and the interview setup. In the following subsections, we explain the three sessions in more detail, followed by the results of each session.

We conducted three study sessions at the three main headquarters of the emergency services across the country, which gave access to large testing areas to set up realistic training missions with the flying drone swarm as well as access to the most experienced drone pilots.

3.1 Session 1: Overview of Drones

For the first session, we focused on visualizing an overview of the drones where the prototype can be seen in Figure 1. This included understanding what the drone operators needed when maintaining an overview of the drones and initial ideas for controlling them. Some of the details in this session were also discussed in our prior work [21]. Two research team members controlled the drones manually, mimicking autonomous behavior until someone used the selection control. We used a Wizard-of-Oz approach for two of the three control methods (beacon and leader drone) - a common practice in drone interaction research [3, 10, 24]. We flew two DJI Mavic Enterprise Advanced Dual drones around an area privately owned by the emergency services. In the demonstration, the participants completed a few tasks, which we designed to allow the participants to consider whether functionalities and the control methods were relevant for potential SAR missions. The demonstration was followed by a co-design session, where the participants expressed their thoughts about the prototype.

3.1.1 Challenges to Current Setup. The drone technologies used are all standard DJI M300 and Mavic 2 Dual drones with the Smart



Figure 1: Study procedure including prototypes and interviews for the initial prototype, advanced swarm control prototype, and interactive video feed prototype.

Controller and Pilot app, which provides manual flight control and simple mapping missions. The participants claimed that the current system works, yet they clarified that the software is limited. This is most apparent in regard to automated flight paths, where the operators cannot define the flight start and end points nor the camera angle. This was a significant limitation since elements such as the terrain and information given by the police significantly impacted where the drone should fly.

Collaboration is also a challenge that emergency services face amidst a SAR operation. P2 noted that helicopters were a challenge when operating drones: “[...] it’s also difficult to work together with the helicopters, which means that there’s no cooperation [...] the drones are grounded.”

However, the main challenge faced in their drone missions is processing the video feed, especially in dense areas where the terrain is filled with different colors, items, and patterns. This is done manually by someone who has to be replaced after ~20 minutes as processing the feed is mentally demanding and can lead to what the emergency services refer to as *image blindness*. Due to this factor, P2 showed skepticism toward a multi-drone system as they said “[...] it doesn’t necessarily add a whole lot of value because for the current situation, it is bringing just a lot more data, and then the

bottleneck will be the people processing the data.” However, he elaborated that he was not against the idea if “[...] there is something to do the process of data on behalf of us [...]”.

3.1.2 Physical Devices. The participants agreed that using a tablet to gather an overview of the drones and view the flight paths was sufficient. Still, the interface should be designed for ‘sausage fingers’ and would be too small to identify objects in the video feed. P1 noted that if a device is meant to be used outdoors, the tablet would need to have “[...] the right surface that can be used in hot weather and rain [...] and also able to work in strong sunlight.” The team often experiences that touch screens do not respond very well when exposed to heat, and the battery quickly drains in cold weather. We asked whether styluses would be a viable option for them. However, P1 said: “We shake our fingers too much with it, it’s simply too cold. [...] a button on the screen is what helps us. Or that the controller has a physical button that has the function for that.”

3.1.3 Control Methods. The participants generally found that the three control methods presented were relevant to SAR efforts. A vital feature they claimed should be the same across the methods is the ability to modify the drones’ behavior to comply with the current situation. They thought it would be valuable if the system

	Session 1 - Overview of drones		Session 2 - Swarm control		Session 3 - Video feed control	
Participant	P1*	P2	P3	P4	P5	P1*
Role	Senior Sergeant	Senior Sergeant	SAR specialist	SAR specialist	Senior Sergeant	Senior Sergeant
Age	42	32	26	50	46	42
Drone Exp. (Years)	6	9	3	1.5	4	6
SAR Exp. (Years)	23	10	2.5	1.5	24	23

Table 1: List of the participants working directly with the drones in SAR missions. *P1 participated twice as he was the nationwide leader for the drone team and the main contact and coordinator for the other study participants.

could automatically suggest beacons based on the terrain. This could help the team identify which areas should be avoided or thoroughly inspected.

The participants stressed that in the final system, it should be very minimal how much the operator should interact with the drones directly. P2 said, *“I think the idea with drone swarms is that we don’t need to control them [...]”*.

3.2 Session 2: Swarm Control

From the first session, we refined the prototype to align with users’ comments more closely. This included implementing the possibility of retrieving the live video feed from the drones.

With the updated prototype, we went to another regional department and focused more on their opinions on the swarm controls. Due to the location, we could not deploy the drones outside. Instead, we opted to deploy four DJI Air 2Ss through the DJI assistant’s simulator, as the prototype would act the same as if the drones were flying in a physical space. Similar approaches have been used in relevant studies [1, 25]. After introducing the prototype’s features, we asked participants to use and explore the application. Finally, we conducted a co-design discussion, with the prototype always visible and working on an iPad in front of them. This allowed our participants to point out and specify what they talked about during the interview.

3.2.1 Drone Behavior. The participants found the selection and beacon control useful for controlling multiple drones but did not find much relevance in the leader drone control method. P3 said *“[...] I don’t find it useful regarding SAR missions[...] I don’t think there needs to be a leader drone; I just think [...] that you want to be able to control more drones at the same time.”* Instead, they found it more relevant to have the drones fly independently in each corner of the search area. They also expressed the importance of visually differentiating the drones from one another, as they wanted to identify and deal with any problems quickly.

3.2.2 Video Feed Layouts. The participants were shown various layouts to display the video feeds. However, there was some skepticism about having multiple video feeds for a single person to review. They estimated that to avoid overwhelming the viewer, there should not be more than four video feeds displayed on the same screen. They welcomed the idea of displaying multiple video feeds, but only if there was an integrated object detection layer that could alleviate data processing tasks from the team.

3.3 Session 3: Video Feed Control

In preparation for the final session, we focused on refining the display of video feeds. Although the control methods were used in the search task, we focused on the interactions with video feeds. We designed layouts for the 11" iPad Pro and the larger LED screen. We flew the swarm of three DJI Air 2Ss autonomously in a mock search mission. However, three research team members always had the physical Smart Controller in hand to override if a problem would arise. The emergency services have a dedicated truck for drone operations, containing all drone equipment and a 30" LED for enlarging the video feed from one active drone. We used the LED screen for the video layout designed for larger screens. We conducted a co-design session after the demonstration. The prototype was running during the session with a large projected screen to show the video layouts. A whiteboard enabled the participants to explain through sketching their visions for future prototypes.

3.3.1 Differentiating the Video Feeds. None of the drone operators we interviewed was colorblind; therefore, they thought that representing the drones and their feed with colors was sufficient as long the colors were clearly visible. However, using colors has the difficulty of not scaling properly with the number of drones, and you would potentially end up in a situation where the colors are too similar. When asked about alternatives using shapes as a unique identifier, they said that they would end up in a similar situation as with the colors, where P6 said: *“[...] you will have nine symbols, where you can stand and discuss, what the (explicit) should I call it [...]?”* The safest identifier, they stated, would be to use numbers to differentiate across a large number of drones.

In the demonstration, the drones flew in a location well-known to the participants, following the same pattern and direction, leading to similar video feeds. The participants said it was relatively easy to maintain an overview of the videos in this scenario. However, it would not be as easy if they flew in different directions, where P1 explained: *“As soon two of them start to turn on the Z-axis while they’re flying then you make greater demands on the brain of the person who stands and looks at it [...] I would imagine that I would feel more pressured if it was not an area where I knew the terrain beforehand.”* In cases where the drones find something, it would be necessary to give time for a person to react and see and accept what they have seen is relevant to the system. They suggested that the drone pause or follow the target while notifying the team until an operator told it otherwise.

3.3.2 Utilizing Multiple Devices and Screens. Extensive discussion was focused on the separation of roles across the different devices.

Using the existing prototype, the iPad could be used as a controller for what was displayed on the larger screen. Here, the participants suggested that in the case of multiple larger screens, it would be interesting to have the option to decide which screen the video should be displayed on instead of having it displayed on all connected screens. This would be helpful if a user has found something that could interest the rest of the team, they could display it for part of the team but not distract other team members focused on other areas of the search.

P1 was adamant in the claim that they would need at least “one big screen, one management tablet” and three additional tablets for view-only access. This would allow people outside the drone team to maintain an overview of what the drones are seeing, so they would also better understand the unfolding situation. These additional tablets, by default, should not control the drones but could act as a failover if needed.

4 DISCUSSION

Although this study focused on potential multi-drone usage in SAR, we will discuss how the results relate to general multi-drone interfaces. Throughout the section, we discuss topics and design considerations that need to be a focus in future multi-drone systems.

4.1 Input Devices for Rugged Operations

Multi-drone systems are bound to be used in an outdoor space, where weather conditions will significantly impact which devices are most suitable for managing a swarm of drones. The participants expressed the necessity to allow controls without relying on precision or gestures. Therefore, physical buttons or larger icon sizes on the interface are valuable. Moreover, extensive gesture controls are also discouraged by some participants as they provide less feedback than physical buttons, and accidental or misinterpreted touches can severely impact the user’s intent. Future interfaces should consider utilizing additional physical buttons or prioritize on-screen buttons that provide the proper size and feedback if the tablet system is used outdoors.

4.2 Size, Number, and Placement of Screens

While tablets have disadvantages in certain weather conditions, the emergency services of Denmark still use iPad Pros as they are easy to handle and portable. However, one participant expressed concern that the more complex the system would become, the harder it would be to navigate around the screen. It is furthermore noteworthy that the SAR van was equipped with a printer for printing out larger physical maps for the team to use for discussions with the incident commander. These factors indicate that extensive multi-drone interfaces might value larger screens for a better overview and control; however, the portability of a tablet is lost and might not be preferred in other situations. Research has also found potential in non-screen approaches for drone-related SAR missions such as mixed reality and wearable technology [4, 6, 18, 29]. Future research could seek to understand how different devices benefit different contexts.

We gained initial insights into how the camera interface should be designed if the goal was to locate an object. Although it depends on the situation, an operator could only observe very few feeds

simultaneously, which other monitoring and surveillance fields also experience [22]. In general, in a context where it is required for a person to constantly keep an eye on them, a multi-drone system is only viable if the system itself can carry some of the workloads with the help of automation, such as auto-detection of objects or anomalies. A person still has to continuously accept or approve whether the automation has found something, and it is still uncertain how a system should be designed to properly display what they have identified and give time for the user to take action. Research on multi-drone systems for observation and monitoring should seek to understand how drones need to behave when an object of interest is found and how the interface should adjust to provide the best possibility to take action.

Collaboration between the different parties in a SAR mission was a challenge as there is a need to tightly coordinate where and how the drones should fly, especially when other entities are in the area. Through our co-design sessions, the participant showed interest in giving parties outside the drone team more insight into the drones’ information. This could be facilitated by incorporating multiple tablets out in the field with different access levels, such as read-only access or access only to one specific drone. Existing research has touched upon the idea of multiple users collaborating to control the drones’ and robots’ behavior to fit everybody’s needs [28, 30]. Future research could help to understand how to support collaboration between drone operators and drone swarms, as well as with other stakeholders that are not an immediate part of the operation.

4.3 Situation Awareness and Trust

Throughout the study, various discussions were focused on understanding the automation and aspects that would impact the user’s SA and trust with autonomous features [13, 31]. When the demonstrations occurred outside, participants were cautious of their surroundings and constantly ensured that the drones were in a stable and safe operating state. However, this is difficult as the number of drones increases, and some drones will be out of the user’s visual line of sight.

For some participants, it was essential to differentiate the drones so they could immediately identify which drone acts abnormally or has found an object. We have attempted to use colors and shapes to address this issue, and while it is sufficient for now, it will have a problem scaling in the future. A numbering system would be ideal, but it might still not be enough if the drones fly in a cluster and overlap. At the moment, all drones involved in the operation are always displayed on the map, but it is possible that it is not always necessary to display them. To safely employ such a feature, studies need to explore if it is necessary to show the individual drone at all times or explore alternatives that scale more appropriately with an increasing number of drones.

To maintain SA, the participants said they would need to know what the drones have done and what they will do. The work of Endsley [14] indicates that for the user to make informed decisions, it is essential not to omit information but make them available in a manner that allows the user to gain it without being overwhelmed or having to seek after it. Like Agrawal et al. [1], research should

seek to understand how we can design interfaces for maintaining a high level of SA.

It was apparent from the co-design sessions that a multi-drone system would only be viable if a layer of automation supported the users. In the case of SAR, object detection in the video feed and flight automation would be critical to the success of drone swarms. However, research shows that visual representation of automation can result in the *overtrust* in a system, where the users will overestimate the capabilities of the system [12]. The users should still be allowed to intervene, not only when a problem occurs but also when the automation is not behaving to their liking. This is the very problem some users have experienced with the current commercial software for single drone control [9, 37]. The appropriate granularity of user control in adjusting the automation is still uncertain. Future research should seek to develop guidelines for how automation should be handled in multi-drone systems.

5 CONCLUSION & FUTURE WORK

In this paper, we sought to establish key challenges and requirements for designing multi-drone interfaces. We selected the search and rescue (SAR) domain as our case, as it provides a realistic and critical scenario for drone swarm usage. We developed three prototype versions of a drone swarm interface to enable the control of physical drones outdoors. From our co-design sessions, we discovered that collaboration, devices, and video feedback analysis were major hurdles for drone usage in SAR. For a multi-drone system to be viable, we address it by coming up with initial requirements as to what future interfaces need to consider. This includes designing responsive and simple input methods that do not rely on gestures, designing for potential collaboration, and allowing different levels of automation to elicit trust in the user. The current prototype was not created to facilitate better situation awareness and trust explicitly but focused on ways of operating and controlling drones. There is still a need to explore more extensively how to design interactions and support user control between operators and drone swarms. We plan to further develop the prototype to conform to the initial design requirements to utilize the initial requirement we gained from our discussion with the emergency services while addressing situation awareness and control as a main component.

ACKNOWLEDGMENTS

We thank Andreh Bassam Bahodi, Rasmus Skov Buchholdt, Andreas Daugbjerg Christensen, Andreas Skjoldgaard Andersen, Shpend Gjela, and Philip Michaelsen for helping the project come to fruition. We thank the Danish emergency management agency professionals and Robotto for their participation. This work is supported by the Innovation Fund Denmark for the project DIREC (9142-00001B).

REFERENCES

- [1] Ankit Agrawal, Sophia J. Abraham, Benjamin Burger, Chichi Christine, Luke Fraser, John M. Hoeksema, Sarah Hwang, Elizabeth Travnik, Shreya Kumar, Walter Scheirer, Jane Cleland-Huang, Michael Vierhauser, Ryan Bauer, and Steve Cox. 2020. The Next Generation of Human-Drone Partnerships: Co-Designing an Emergency Response System. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, 1–13. <https://doi.org/10.1145/3313831.3376825>
- [2] Ankit Agrawal and Jane Cleland-Huang. 2021. Explaining Autonomous Decisions in Swarms of Human-on-the-Loop Small Unmanned Aerial Systems. 9 (2021), 15–26. <https://doi.org/10.1609/hcomp.v9i1.18936>
- [3] Majed Al Zayer, Sam Tregillus, Jiwan Bhandari, Dave Feil-Seifer, and Eelke Folmer. 2016. Exploring the Use of a Drone to Guide Blind Runners. In *Proceedings of the 18th International ACM SIGACCESS Conference on Computers and Accessibility*. ACM, 263–264. <https://doi.org/10.1145/2982142.2982204>
- [4] Sultan A. Alharthi, Nick LaLone, Ahmed S. Khalaf, Ruth Torres, Lennart Nacke, Igor Dolgov, and Zachary O. Toups. 2018. Practical Insights into the Design of Future Disaster Response Training Simulations. (2018). <https://par.nsf.gov/biblio/10061145-practical-insights-design-future-disaster-response-training-simulations>
- [5] Sultan A. Alharthi, Nicolas James LaLone, Hitesh Nidhi Sharma, Igor Dolgov, and Z. O. Toups. 2021. An Activity Theory Analysis of Search & Rescue Collective Sensemaking and Planning Practices. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM, 1–20. <https://doi.org/10.1145/3411764.3445272>
- [6] Sultan A. Alharthi, Hitesh Nidhi Sharma, Sachin Sunka, Igor Dolgov, and Zachary O. Toups. 2018. Designing Future Disaster Response Team Wearables from a Grounding in Practice. In *Proceedings of the Technology, Mind, and Society*. ACM, 1–6. <https://doi.org/10.1145/3183654.3183662>
- [7] Muhammad Yeasir Arafat and Sangman Moh. 2019. Routing Protocols for Unmanned Aerial Vehicle Networks: A Survey. 7 (2019), 99694–99720. <https://doi.org/10.1109/ACCESS.2019.2930813> Conference Name: IEEE Access.
- [8] Oscar Bjurling, Rego Granlund, Jens Alfredson, Mattias Arvola, and Tom Ziemke. 2020. Drone Swarms in Forest Firefighting: A Local Development Case Study of Multi-Level Human-Swarm Interaction. In *Proceedings of the 11th Nordic Conference on Human-Computer Interaction: Shaping Experiences, Shaping Society*. ACM, Tallinn Estonia, 1–7. <https://doi.org/10.1145/3419249.3421239>
- [9] Gloria L. Calhoun, Mark H. Draper, and Heath A. Ruff. 2009. Effect of Level of Automation on Unmanned Aerial Vehicle Routing Task. 53, 4 (2009), 197–201. <https://doi.org/10.1177/154193120903500408>
- [10] Jessica R. Cauchard, Jane L. E, Kevin Y. Zhai, and James A. Landay. 2015. Drone & me: an exploration into natural human-drone interaction. In *Proceedings of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '15)*. Association for Computing Machinery, 361–365. <https://doi.org/10.1145/2750858.2805823>
- [11] Serge Chaumette and Frédéric Guinand. 2017. Control of a Remote Swarm of Drones/Robots Through a Local (Possibly Model) Swarm: Qualitative and Quantitative Issues. In *Proceedings of the 14th ACM Symposium on Performance Evaluation of Wireless Ad Hoc, Sensor, & Ubiquitous Networks*. ACM, 41–45. <https://doi.org/10.1145/3134829.3134840>
- [12] Mark Colley, Max Rädler, Jonas Glimmann, and Enrico Rukzio. 2022. Effects of Scene Detection, Scene Prediction, and Maneuver Planning Visualizations on Trust, Situation Awareness, and Cognitive Load in Highly Automated Vehicles. 6, 2 (2022), 1–21. <https://doi.org/10.1145/3534609>
- [13] Stephen R. Dixon and Christopher D. Wickens. 2006. Automation Reliability in Unmanned Aerial Vehicle Control: A Reliance-Compliance Model of Automation Dependence in High Workload. 48, 3 (2006), 474–486. <https://doi.org/10.1518/001872006778606822>
- [14] Mica R Endsley. 2016. *Designing for Situation Awareness: An Approach to User-Centered Design*. CRC Press.
- [15] Mica R. Endsley and Mark D. Rodgers. 1994. Situation Awareness Information Requirements Analysis for En Route Air Traffic Control. 38, 1 (1994), 71–75. <https://doi.org/10.1177/154193129403800113>
- [16] Stefan Feuerriegel, Rúben Geraldes, Artur Gonçalves, Ziqi Liu, and Helmut Prendinger. 2021. Interface Design for Human-Machine Collaborations in Drone Management. 9 (2021), 107462–107475. <https://doi.org/10.1109/ACCESS.2021.3100712>
- [17] Eitan Frachtenberg. 2019. Practical Drone Delivery. 52, 12 (2019), 53–57. <https://doi.org/10.1109/MC.2019.2942290>
- [18] Marlana R. Fraune, Ahmed S. Khalaf, Mahlet Zemedie, Poom Pianpak, Zahra NaminiMianji, Sultan A. Alharthi, Igor Dolgov, Bill Hamilton, Son Tran, and Z. O. Toups. 2021. Developing Future Wearable Interfaces for Human-Drone Teams through a Virtual Drone Search Game. 147 (2021), 102573. <https://doi.org/10.1016/j.jhcs.2020.102573>
- [19] Sharad Kumar Gupta and Dericks P. Shukla. 2018. Application of drone for landslide mapping, dimension estimation and its 3D reconstruction. 46, 6 (2018), 903–914. <https://doi.org/10.1007/s12524-017-0727-1>
- [20] Viviane Herdel, Lee J. Yamin, and Jessica R. Cauchard. 2022. Above and Beyond: A Scoping Review of Domains and Applications for Human-Drone Interaction. In *CHI Conference on Human Factors in Computing Systems (2022-04-29) (CHI '22)*. Association for Computing Machinery, 1–22. <https://doi.org/10.1145/3491102.3501881>
- [21] Maria-Theresa Oanh Hoang, Niels van Berkel, Mikael B Skov, and Timothy Merriitt. 2022. Challenges Arising in a Multi-Drone System for Search and Rescue. In *Adjunct Proceedings of the 12th Nordic Conference on Human-Computer Interaction*. 1–5.
- [22] Helen M. Hodgetts, François Vachon, Cindy Chamberland, and Sébastien Tremblay. 2017. See No Evil: Cognitive Challenges of Security Surveillance and Monitoring. 6, 3 (2017), 230–243. <https://doi.org/10.1016/j.jarmac.2017.05.001>

- [23] Aya Hussein and Hussein Abbass. 2018. Mixed Initiative Systems for Human-Swarm Interaction: Opportunities and Challenges. In *2018 2nd Annual Systems Modelling Conference (SMC)*. 1–8. <https://doi.org/10.1109/SYSMC.2018.8509744>
- [24] Kari Daniel Karjalainen, Anna Elisabeth Sofia Romell, Photchara Ratsamee, Asim Evren Yantac, Morten Fjeld, and Mohammad Obaid. 2017. Social Drone Companion for the Home Environment: a User-Centric Exploration. In *Proceedings of the 5th International Conference on Human Agent Interaction*. ACM, 89–96. <https://doi.org/10.1145/3125739.3125774>
- [25] Ahmed S. Khalaf, Poom Pianpak, Sultan A. Alharthi, Zahra NaminiMianji, Ruth Torres, Son Tran, Igor Dolgov, and Zachary O. Toups. 2018. An Architecture for Simulating Drones in Mixed Reality Games to Explore Future Search and Rescue Scenarios. (2018). <https://par.nsf.gov/biblio/10061144-architecture-simulating-drones-mixed-reality-games-explore-future-search-rescue-scenarios>
- [26] Andreas Kolling, Katia Sycara, Steve Nunnally, and Michael Lewis. 2013. Human Swarm Interaction: An Experimental Study of Two Types of Interaction with Foraging Swarms. *Journal of Human-Robot Interaction* 2, 2 (2013), 103–128. <https://doi.org/10.5898/JHRI.2.2.Kolling>
- [27] Andreas Kolling, Phillip Walker, Nilanjan Chakraborty, Katia Sycara, and Michael Lewis. 2016. Human Interaction With Robot Swarms: A Survey. 46, 1 (2016), 9–26. <https://doi.org/10.1109/THMS.2015.2480801>
- [28] G. J. M. Kruijff, M. Janíček, S. Keshavdas, B. Larochelle, H. Zender, N. J. J. M. Smets, T. Mioch, M. A. Neerincx, J. V. Diggelen, F. Colas, M. Liu, F. Pomerleau, R. Siegart, V. Hlaváč, T. Svoboda, T. Petříček, M. Reinstein, K. Zimmermann, F. Pirri, M. Gianni, P. Papadakis, A. Sinha, P. Balmer, N. Tomatis, R. Worst, T. Linder, H. Surmann, V. Tretyakov, S. Corrao, S. Pratzler-Wanczura, and M. Sulk. 2014. Experience in System Design for Human-Robot Teaming in Urban Search and Rescue. In *Field and Service Robotics: Results of the 8th International Conference*, Kazuya Yoshida and Satoshi Tadokoro (Eds.). Springer, 111–125. https://doi.org/10.1007/978-3-642-40686-7_8
- [29] Nicolas LaLone, Sultan A. Alharthi, and Z O. Toups. 2019. A Vision of Augmented Reality for Urban Search and Rescue. In *Proceedings of the Halfway to the Future Symposium 2019*. ACM, 1–4. <https://doi.org/10.1145/3363384.3363466>
- [30] Jiannan Li, Mauricio Sousa, Ravin Balakrishnan, and Tovi Grossman. 2021. Constellation: a Multi-User Interface for Remote Drone Tours. In *Proceedings of the 9th International Conference on Human-Agent Interaction*. ACM, 277–282. <https://doi.org/10.1145/3472307.3484685>
- [31] Martin Lochner, Andreas Duenser, and Shouvojit Sarker. 2020. Trust and Cognitive Load in semi-automated UAV operation. In *Proceedings of the 31st Australian Conference on Human-Computer-Interaction (OZCHI'19)*. Association for Computing Machinery, 437–441. <https://doi.org/10.1145/3369457.3369509>
- [32] Michael J Muller and Allison Druin. 2012. Participatory design: The third space in human-computer interaction. In *The Human-Computer Interaction Handbook*. CRC Press, 1125–1153.
- [33] Wilson O. Quesada, Jonathan I. Rodriguez, Juan C. Murillo, Gustavo A. Cardona, David Yanguas-Rojas, Luis G. Jaimes, and Juan M. Calderón. 2018. Leader-Follower Formation for UAV Robot Swarm Based on Fuzzy Logic Theory. In *Artificial Intelligence and Soft Computing (Lecture Notes in Computer Science)*, Leszek Rutkowski, Rafal Scherer, Marcin Korytkowski, Witold Pedrycz, Ryszard Tadeusiewicz, and Jacek M. Zurada (Eds.). Springer International Publishing, 740–751. https://doi.org/10.1007/978-3-319-91262-2_65
- [34] Davide Di Ruscio, Ivano Malavolta, Patrizio Pelliccione, and Massimo Tivoli. 2016. Automatic generation of detailed flight plans from high-level mission descriptions. In *Proceedings of the ACM/IEEE 19th International Conference on Model Driven Engineering Languages and Systems*. ACM, 45–55. <https://doi.org/10.1145/2976767.2976794>
- [35] Fabrice Saffre, Hanno Hildmann, and Hannu Karvonen. 2021. The Design Challenges of Drone Swarm Control. In *Engineering Psychology and Cognitive Ergonomics*, Don Harris and Wen-Chin Li (Eds.). Vol. 12767. Springer International Publishing, 408–426. https://doi.org/10.1007/978-3-030-77932-0_32
- [36] EBN Sanders and PJ Stappers. 2012. *Convivial toolbox: Generative research for the front end of design*. BIS.
- [37] Ben Shneiderman. 2022. *Human-Centered AI*. Oxford University Press.
- [38] Ryotaro Temma, Kazuki Takashima, Kazuyuki Fujita, Koh Sueda, and Yoshifumi Kitamura. 2019. Third-Person Piloting: Increasing Situational Awareness using a Spatially Coupled Second Drone. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology*. ACM, 507–519. <https://doi.org/10.1145/3332165.3347953>
- [39] Pietro Tosato, Daniele Facinelli, Maurizio Prada, Luca Gemma, Maurizio Rossi, and Davide Brunelli. 2029. An Autonomous Swarm of Drones for Industrial Gas Sensing Applications. In *2019 IEEE 20th International Symposium on "A World of Wireless, Mobile and Multimedia Networks" (WoWMoM)*. 1–6. <https://doi.org/10.1109/WoWMoM.2019.8793043>
- [40] Yu-Hsuan Tu, Stuart Phinn, Kasper Johansen, Andrew Robson, and Dan Wu. 2020. Optimising drone flight planning for measuring horticultural tree crop structure. 160 (2020), 83–96. <https://doi.org/10.1016/j.isprsjprs.2019.12.006>
- [41] Daniele Ventura, Michele Bruno, Giovanna Jona Lasinio, Andrea Belluscio, and Giandomenico Ardizzone. 2016. A low-cost drone based application for identifying and mapping of coastal fish nursery grounds. 171 (2016), 85–98. <https://doi.org/10.1016/j.ecss.2016.01.030>
- [42] Markus Waibel, Bill Keays, and Federico Augugliaro. 2017. Drone shows: Creative potential and best practices. , 18 pages. <https://doi.org/10.3929/ETHZ-A-010831954>

6 APPENDIX A: PROTOTYPE SCREEN CAPTURES

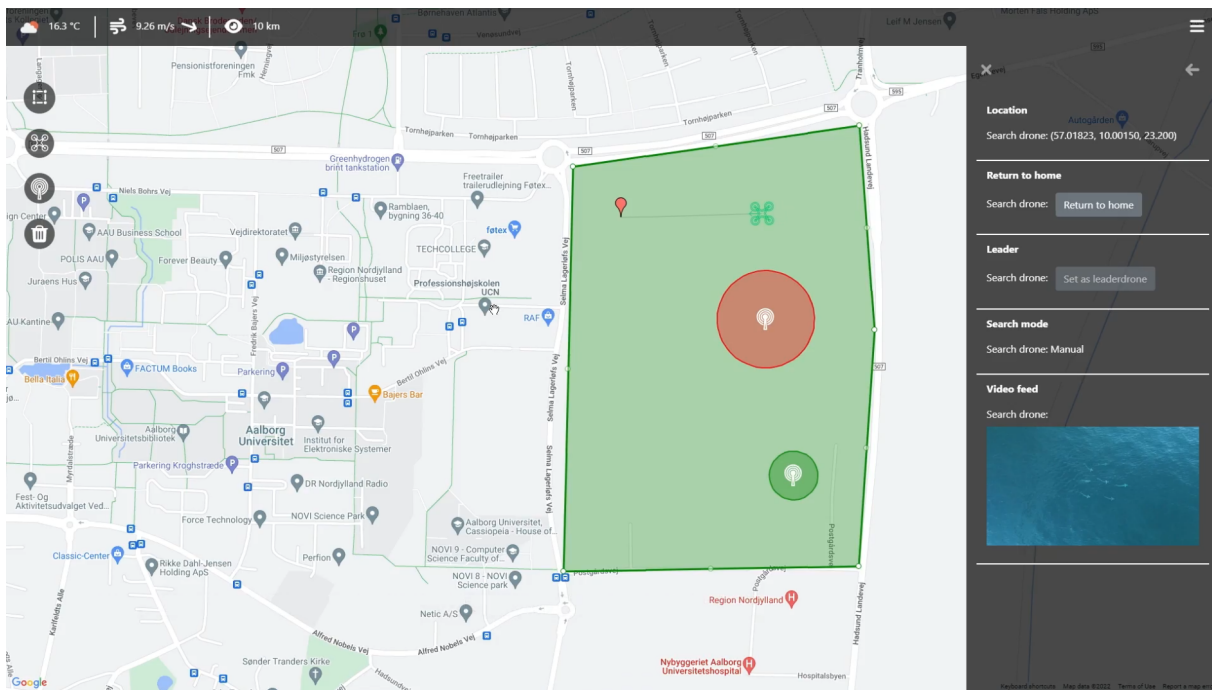


Figure 2: Prototype from Session 1: Overview of Drones.

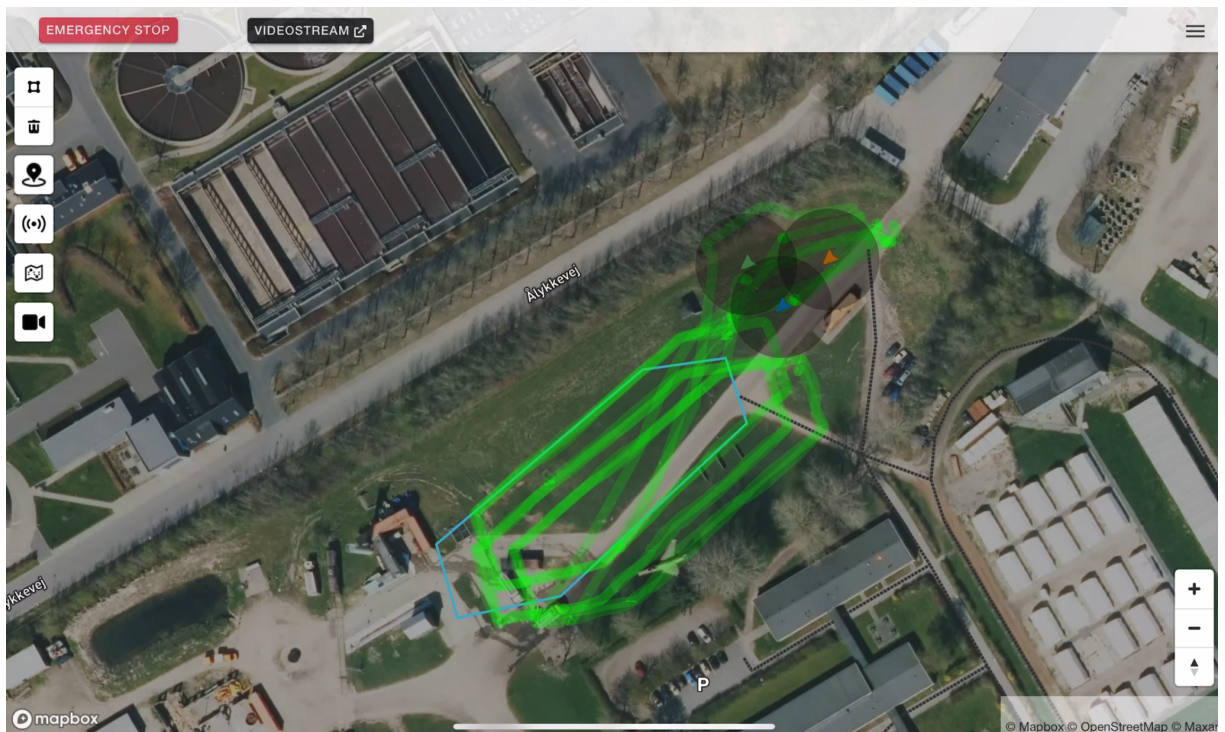


Figure 3: Prototype from Session 2: Swarm Control & Session 3: Video Feed Control.

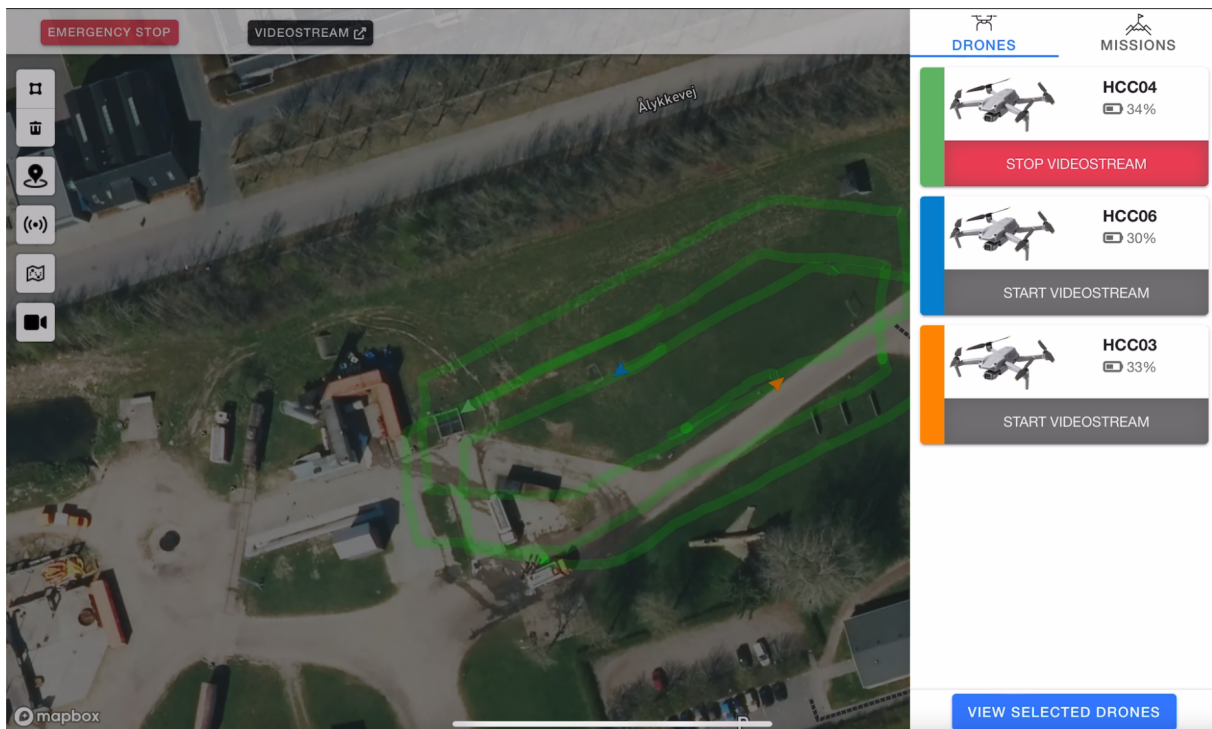


Figure 4: Drone side menu of the prototype from Session 2: Swarm Control & Session 3: Video Feed Control.

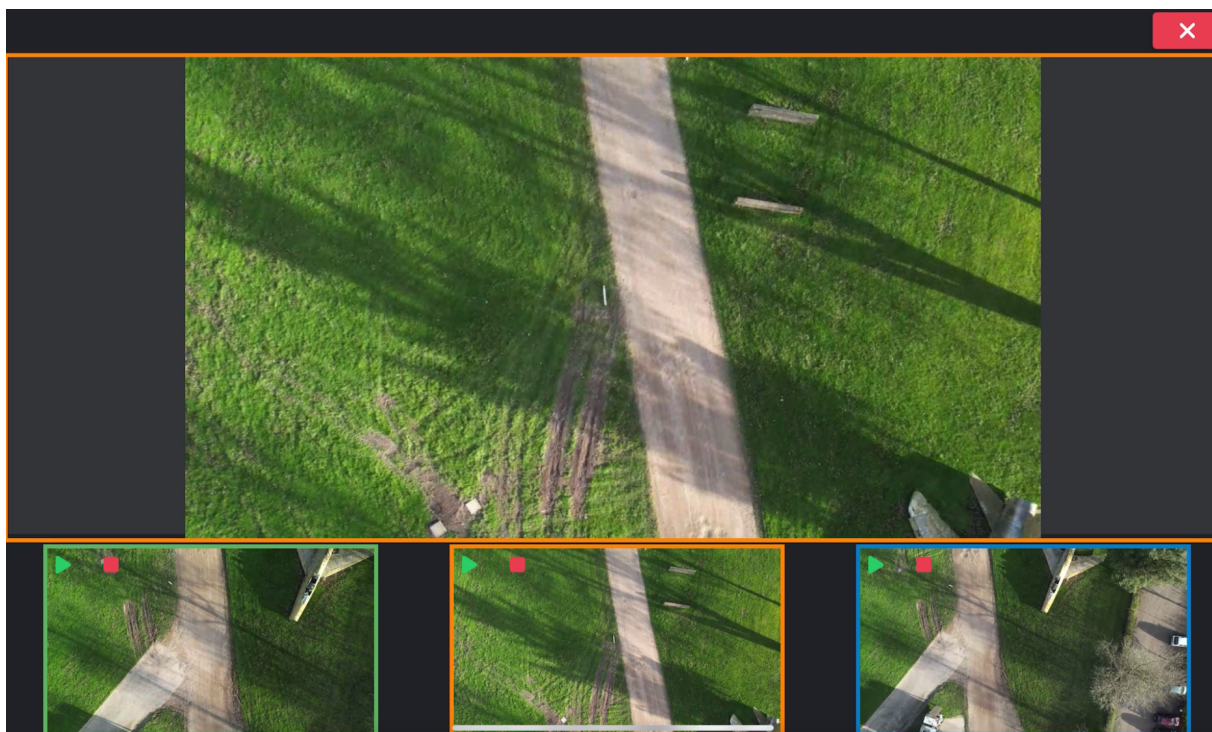


Figure 5: Video feed layout from Session 3: Video Feed Control.